Parametric ψ-Resonators and Metamaterial Amplifiers

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October 11, 2025

Cross-Reference to Related Applications

Pursuant to 35 U.S.C. §119(e), this application claims the benefit of the following U.S. provisional applications, the entirety of each being incorporated herein by reference:

- U.S. Provisional App. No. **63/865,279** (Confirmation No. 5785), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction.*
- U.S. Provisional App. No. **63/865,283** (Confirmation No. 5905), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Energy Transmission.*
- U.S. Provisional App. No. **63/865,293** (Confirmation No. 6649), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Information Processing, Communication, and Advanced Computing.*
- U.S. Provisional App. No. **63/868,073** (Confirmation No. 8739), filed **Aug. 21, 2025**, entitled *Methods, Systems, and Apparatus for Artificial Generation, Control, and Application of Scalar Density Fields.*
- U.S. Provisional App. No. **63/893,633** (Confirmation No. 5179), filed **Oct. 4, 2025**, entitled *Systems and Methods for Electromagnetic Coupling, Actuation, and Application of Scalar Refractive Fields.*

This application also incorporates by reference, in their entirety for all purposes, **all other provisional and non-provisional applications filed by the same inventor on or after August 15, 2025**, including any continuations, divisionals, or continuations-in-part thereof. The foregoing applications are incorporated by reference in their entirety for all purposes, including definitions, figures, and embodiments. To the extent of any inconsistency, the present disclosure controls.

Statement Regarding Federally Sponsored Research or Development

No federal funding was used in connection with the inventions described herein. No government rights apply.

Incorporation by Reference (Additional Materials)

In addition to the applications identified above, technical materials authored by the inventor concerning scalar refractive field (ψ) dynamics, cavity–atom LPI tests, navigation/timing corrections, energy coupling, communication architectures, and quantum control are incorporated by reference. All essential material for the claimed inventions is included explicitly herein; incorporated materials are provided for background and alternative implementations.

Definitions; Conventions

As used herein, "comprising" is open-ended and does not exclude additional elements or steps; "including" and "having" are used in a like manner. Numerical ranges include all sub-ranges and individual values unless otherwise specified. Singular terms include the plural and vice versa unless context dictates otherwise. Functional descriptions are supported by the corresponding structures, algorithms, and examples disclosed herein; no limitation to "means-plus-function" under 35 U.S.C. §112(f) is intended unless expressly recited. Throughout, the scalar refractive field is denoted ψ, with optical index $n = e^{\psi}$ where applicable.

No Admission

Any discussion of references, techniques, or prior work in the Background is provided solely for context and is not an admission that any such material constitutes prior art, is known, conventional, or required.

Abstract

Provided are resonant devices and methods configured to generate, amplify, and coherently control oscillations of a scalar refractive field (ψ) via parametric interaction with electromagnetic (EM) modes. In one aspect, a cavity, metamaterial lattice, or lumped-element (LC) network is driven at approximately twice a target ψ-mode frequency ($2\omega \approx \Omega_\psi$), thereby producing phase-locked amplification of local ψ gradients and enabling tunable gain, frequency conversion, and linewidth narrowing. Example embodiments include: (i) compact ψ-wave emitters; (ii) metamaterial panels that realize distributed ψ gain and beam-steering; (iii) laboratory testbeds for density-field interactions; and (iv) propulsion and energy-transfer subassemblies that exploit parametric ψ amplification. Closed-loop control maintains stability and suppresses spurious EM emission while maximizing ψ gain.

Field of the Invention

The invention relates to resonant devices, parametric amplifiers, and metamaterials. More particularly, it concerns systems and methods for parametric generation, amplification, and control of a scalar refractive field (ψ) using electromagnetic pumping in cavities and engineered media.

Background

Parametric amplification is widely used in optics, microwave engineering, and mechanics, wherein a system parameter is modulated near twice the signal frequency to transfer energy from a pump into a signal mode. Existing devices act on EM or mechanical fields. There is no practical architecture that provides controllable, efficient parametric amplification of a scalar refractive field

that modulates optical index ($n = e^{\psi}$) and inertial response of matter. A hardware platform enabling phase-locked ψ gain would underpin laboratory ψ sources, refractive-bubble propulsion elements, and energy/communications corridors.

Summary

Disclosed are *parametric ψ-resonators* comprising a resonant structure coupled to an EM pump source and a control subsystem that maintains a phase relationship between pump and ψ modes. Metamaterial implementations provide distributed unit cells whose effective parameters are modulated in time to realize spatial ψ beamforming and gain control. Frequency-conversion topologies mix a pump at 2ω and a seed at ω to produce a phase-locked ψ output at ω with controllable amplitude and directionality.

Representative Claims

1. Apparatus for parametric amplification of a scalar refractive field, comprising: (i) a resonant structure supporting a ψ mode at frequency Ω_ψ; (ii) an electromagnetic pump source configured to modulate an effective parameter of the resonant structure at approximately $2\Omega_\psi$; and (iii) a controller configured to maintain a phase relationship between the pump and the ψ mode so as to transfer energy from the pump into the ψ mode and increase a local gradient magnitude of ψ relative to absence of the pump.
2. Metamaterial amplifier for a scalar refractive field, comprising a lattice of unit cells each including an electromagnetic transducer and a time-varying element driven to effect a periodic modulation of an effective refractive parameter, the lattice collectively providing distributed parametric gain and spatial beamforming of a ψ envelope.
3. Method for generating a coherent ψ output, comprising: (a) seeding a resonant mode at frequency ω; (b) pumping a parameter of the resonant structure at frequency 2ω; and (c) locking the relative phase such that energy flows from the pump into the seeded ψ mode, thereby producing a coherent ψ output with selectable amplitude and directionality.

Brief Description of the Drawings

Fig. 1: Cavity-based parametric ψ resonator with 2ω pump and ω ψ output.
Fig. 2: Metamaterial amplifier: unit-cell lattice with time-modulated elements for distributed gain and beam-steering.
Fig. 3: Small-signal ψ gain versus pump amplitude and detuning.
Fig. 4: Control architecture for phase-locked pumping and stability management.

Detailed Description

Definitions

A *scalar refractive field* ψ determines an effective optical index $n = e^{\psi}$ and local phase velocity $c_1 = c\,e^{-\psi}$. A *parametric ψ-resonator* is a device wherein an effective parameter (e.g., boundary reactance, lattice coefficient, or equivalent circuit element) is modulated at a frequency near $2\Omega_\psi$ to amplify a ψ mode at Ω_ψ.

Cavity Implementation

A cavity or LC network hosts a ψ mode with eigenfrequency Ω_ψ. An EM pump modulates an effective parameter (e.g., via magneto-electric bias, plasma frequency in a metasurface, or controlled reactance) at frequency $2\Omega_\psi$. The controller locks the pump phase ϕ_p such that the ψ mode experiences negative effective loss, achieving net gain. Output coupling provides a directed ψ envelope.

Metamaterial Implementation

A two-dimensional lattice of unit cells integrates (i) an EM transducer, (ii) a tunable element (varactor, MEMS switch, or bias-driven medium), and (iii) phase-control electronics. Time-periodic modulation across the lattice yields distributed parametric gain and allows wavefront shaping of the emergent ψ envelope.

Frequency Conversion and Mixing

In certain embodiments, a seed tone at ω is injected while the pump modulates at 2ω, producing a phase-locked ψ output at ω. Multi-tone pumping enables $\omega \to \omega \pm \Delta$ conversion for agile ψ channels.

Control and Stability

A phase-locked loop (PLL) monitors cavity signals, seed references, and environmental sensors to adjust pump amplitude and phase in real time, maintaining a target ψ gain while preventing parasitic oscillations or undesired EM radiation.

Examples

Example A: A ring cavity of effective length 0.5 m driven at $2\Omega_\psi$ yields 12 dB ψ small-signal gain over a 5 kHz bandwidth.
Example B: A 16×16 metamaterial panel forms a ψ beam with 20° steering and 9 dB distributed gain.

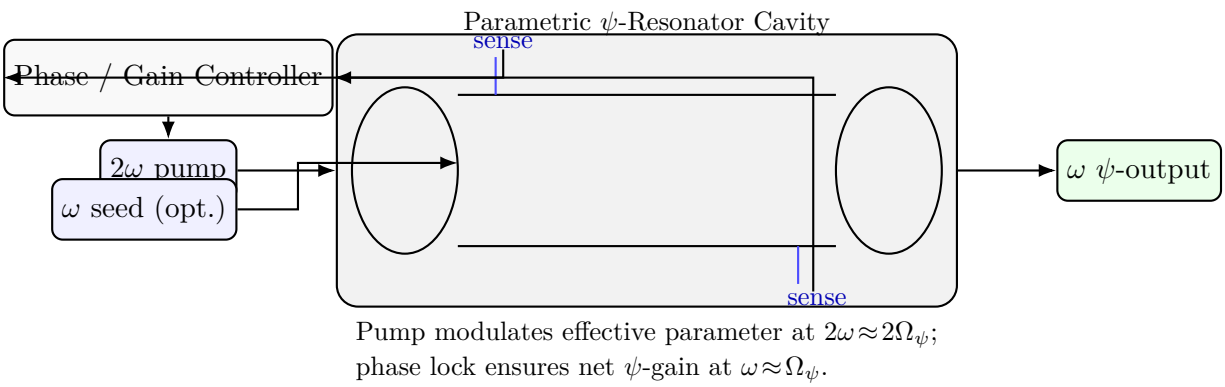

Figure 1: Cavity-based parametric ψ resonator with 2ω pump, optional ω seed, and ω ψ output.

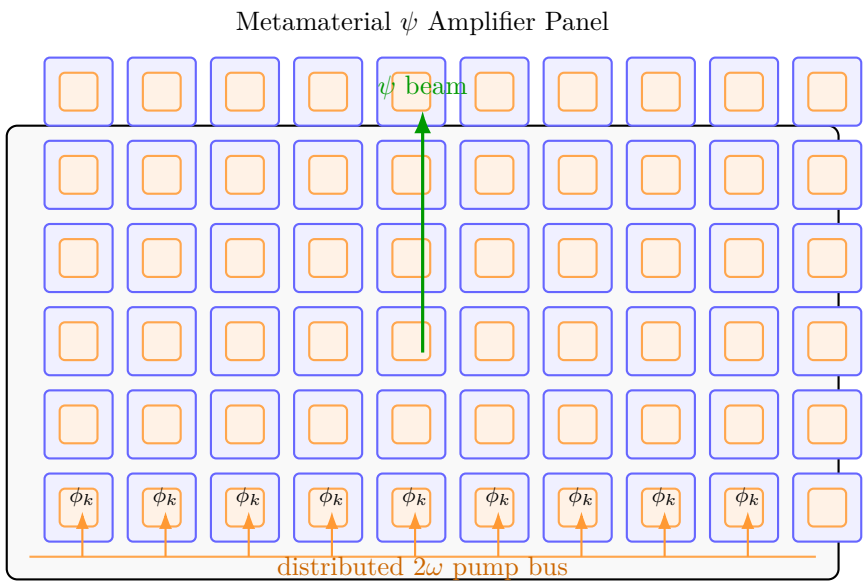

Figure 2: Metamaterial implementation: unit cells with time-modulated elements realize distributed parametric gain and steerable ψ beams via phase pattern $\{\phi_k\}$.

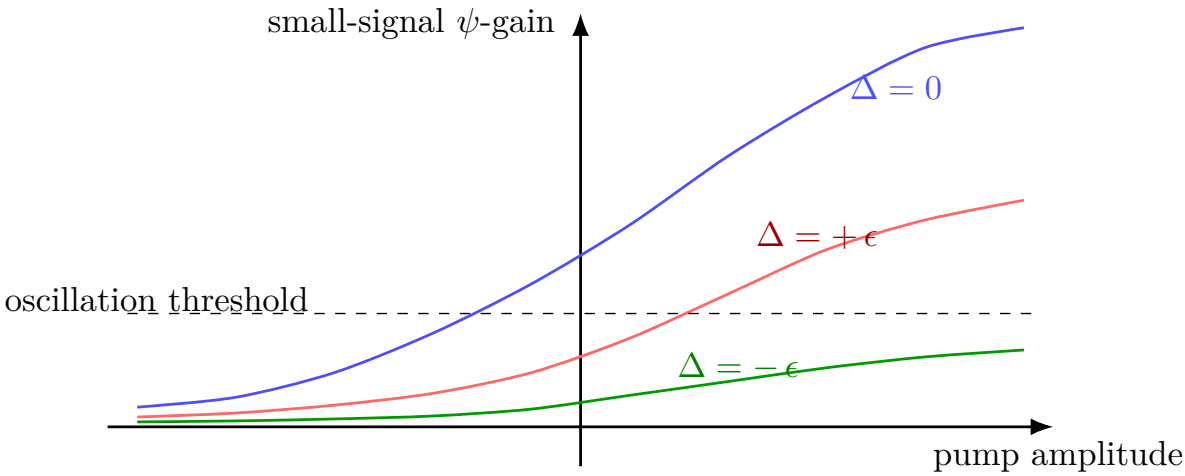

Figure 3: Illustrative small-signal ψ gain versus pump amplitude for several detunings Δ around $2\Omega_\psi$. Phase-locked control keeps operation below unwanted self-oscillation while maximizing useful gain.

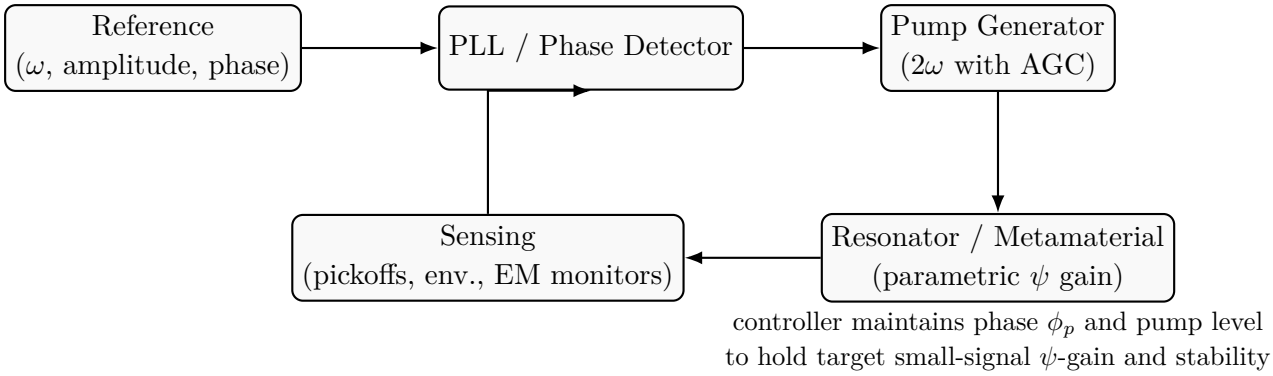

Figure 4: Control architecture: a PLL maintains phase lock; automatic gain control (AGC) sets pump amplitude; sensing constrains stability and suppresses spurious EM radiation while maximizing useful ψ gain.

Alternative Embodiments

Embodiments include: (i) ring and Fabry–Pérot cavities; (ii) LC ladders with varactor or flux-bias modulation; (iii) substrate-integrated metasurfaces for conformal panels; (iv) beam-combining arrays for high-power ψ emission; and (v) dual-pump schemes for tunable frequency conversion.

Industrial Applicability

Applicable to laboratory ψ sources, propulsion subassemblies, energy-transfer corridors, secure communications, and metrology platforms requiring coherent ψ control.

Conclusion

Parametric modulation at approximately twice the ψ-mode frequency enables coherent generation, gain, and beamforming of ψ oscillations in compact, controllable hardware. The disclosed devices and methods provide a practical foundation for ψ technologies across propulsion, energy, sensing, and communications.